



Materials Performance and Characterization

R. Sunder¹

DOI: 10.1520/MPC20140037

Characterization of
Threshold Stress Intensity
as a Function
of Near-Tip Residual
Stress: Theory, Experi-
ment, and Applications

VOL. 4 / NO. 2 / 2015

R. Sunder¹

Characterization of Threshold Stress Intensity as a Function of Near-Tip Residual Stress: Theory, Experiment, and Applications

Reference

Sunder, R., "Characterization of Threshold Stress Intensity as a Function of Near-Tip Residual Stress: Theory, Experiment, and Applications," *Materials Performance and Characterization*, Vol. 4, No. 2, 2015, pp. 105–130, doi:10.1520/MPC20140037. ISSN 2165-3992

ABSTRACT

Threshold stress intensity, ΔK_{th} , is not a material constant, but a variable that is extremely sensitive to load history. This effect has nothing to do with crack closure. It is associated with the influence of near-tip residual stress, σ^* , on instantaneous resistance of crack tip surface layers to failure under atmospheric conditions. An analytical procedure is proposed to estimate σ^* under constant amplitude loading as well as after tensile or compressive overloads. An experimental procedure is proposed to determine ΔK_{th} under these loading conditions. ΔK_{th} estimates obtained under a variety of overloads and underloads as well as under constant amplitude loading show an excellent correlation with computed σ^* . The proposed experimental procedure and analytical relationships appear suitable for adoption as standard practice for an engineering application of threshold stress intensity under variable amplitude loading.

Keywords

threshold stress intensity, residual stress, crack closure, variable-amplitude loading

Manuscript received March 9, 2014; accepted for publication October 6, 2014; published online October 31, 2014.

¹ BiSS (P) Ltd., 497E 14th Cross, 4th Phase, Peenya Industrial Area, Bangalore 560058, India, e-mail: rs@biss.in

Introduction

Service load conditions induce load interaction effects. Prevailing standard laboratory practice to determine threshold stress intensity, ΔK_{th} , is specifically designed to preclude them. It follows that ΔK_{th} , determined using standard practices such as ASTM E647-13 [1] may be unsuitable for engineering application.

A variety of experiments under specially designed programmed load sequences offer irrefutable, quantifiable, and reproducible fractographic evidence that atmospheric near-threshold crack growth rates exhibit strong *cycle sequence sensitivity* to load history² [2–6]. The Elber [7], Wheeler [8], and Willenborg [9] models are incapable of simulating such cycle-by-cycle sensitivity because their transient functions are related to crack extension³. These models focus on the monotonic overload plastic zone that essentially sees cyclic *elastic* response. They ignore cyclic plastic zone response, where stress will be cycle-sequence sensitive even in the absence of crack extension. Cycle-sequence sensitivity is associated with the influence of local stress on resistance to crack extension. It has nothing to do with the phenomenon of fatigue crack closure. On the contrary, only an open crack can induce stress variation in the cyclic plastic zone.

Classical education in fracture mechanics promotes the picture of elastic, ideally-plastic response and the associated crack tip cyclic stress ratio of $R = -1$, quite independent of applied stress intensity, stress ratio, and therefore, load history. Such a picture, driven by the understanding of the crack tip as a singularity and associated infinite crack-tip stress concentration [10], may dull one into the belief that near-tip residual stress, σ^* , is load history independent, *even* under near-threshold conditions. When viewed in a historical context, this should not come as a surprise, given the stereotype understanding of metal fatigue. Wohler and Bauschinger [11,12] may have established the extreme sensitivity of fatigue limit to minimum stress about one and half centuries ago. Yet, the focus of classical education in metal fatigue is on those mechanisms and models that are amenable to ease of illustration, modeling, and experimental measurement such as dislocation theory, reversed slip, and cyclic plastic deformation. In the process, a glaring disconnect appears to have remained unnoticed: while reversed cyclic slip is theoretically *insensitive* to mean stress⁴, fatigue life, particularly, fatigue limit, is extremely *sensitive* to the mean stress component. This dichotomy has not attracted much attention over the years⁵. Elber's discovery of crack closure may have served as a "distraction" in this regard, given the belief that stress ratio as well as many load interaction effects appear to be readily explained by closure.

The role of moisture in controlling atmospheric near-threshold crack growth rates is well known [15]. However, it was the discovery that such environmental

²The phenomenon disappears in high vacuum.

³A noteworthy exception is the possibility of instantaneous *reduction* in crack closure either due to a tensile or compressive overload. Instantaneous *increase* in crack closure is implausible.

⁴Cyclic elasto-plastic strain amplitude is a unique function of stress amplitude. Mean stress does not figure in the Ramberg-Osgood equation [13] adapted to describe cyclic strain amplitude.

⁵Only Manson's citation comes to mind, from a review paper in 1988 [14], suggesting, that a "notable development" for the next twenty years would be an understanding of why fatigue is sensitive to mean stress.

action may be controlled by σ^* that led up to the brittle micro fracture (BMF) model of atmospheric near-threshold crack extension [16]. This opened up the possibility of modeling cycle sequence sensitivity of ΔK_{th} from first principles [17]⁶.

According to the BMF model [16], crack tip surface atomic layers may be embrittled by hydrogen released by reaction of nascent surface with moisture to form oxide and hydroxide. Such activity can cause microscopic crack extension through BMF of surface layers, well before the onset of conditions leading to crack extension by slip. This is a surface diffusion driven phenomenon [18] whose intensity is determined by near-tip stress, σ^* , at the *commencement* of the rising half of the load cycle [16]. As σ^* is a measure of residual elastic strain and therefore, of initial interatomic spacing of atoms in the surface layers, further increase in their spacing with rising load will determine the extent of adsorption of active species [19], and thereby, the propensity for fracture of the surface layers. If local residual stresses are highly compressive, adsorption of active species will be inhibited, thereby allowing the crack-tip to merely stretch as in high vacuum (or, as in the case of a smooth specimen), eventually leading to crack extension by slip, but at ΔK exceeding constant amplitude atmospheric ΔK_{th} .

An important conclusion from the foregoing submission is that near-tip stress at minimum load will determine ΔK_{th} for the next rising load half-cycle. This goes against conventional wisdom relating either mean or maximum stress (or, stress intensity) to ΔK_{th} , but actually makes more sense considering that at the commencement of a load cycle, the crack tip “does not know” the magnitude of the forthcoming load excursion. The state of crack-tip surface layers at the commencement of the loading half-cycle will determine the threshold K -excursion required for onset of crack extension. It follows that there *may* exist, a unique relationship between ΔK_{th} and σ^* . As σ^* will be cycle-sequence sensitive, ΔK_{th} must obviously also be re-computed for each rising half-cycle of loading under a variable amplitude sequence.

The potential for vastly improved modeling of variable-amplitude fatigue by accounting for the synergy of crack closure and history sensitive variation in near-threshold resistance to atmospheric fatigue was underscored in previous work [17]. This was not pursued earlier for lack of clarity as to how exactly to characterize sequence sensitive σ^* and more importantly, for lack of a clearly defined, simple and reproducible experimental procedure to determine ΔK_{th} as a function of tightly controlled σ^* . These were the focus of the present study.

This paper is divided into three parts. The first part describes a simple analytical approach to compute cycle-sequence sensitive σ^* . It extends well-known local stress strain (LSS) approach in notch fatigue analysis to come up with a simple process to estimate σ^* under baseline loading after possible overload/underload sequences. The second part describes an experimental procedure to determine ΔK_{th} under a variety of highly controlled overload conditions. It is designed to enable enforcement of a wide range of highly controlled and reproducible values of σ^* . The third part describes the correlation between experimentally determined ΔK_{th} and computed

⁶Unless otherwise mentioned, this and all subsequent reference to ΔK_{th} imply closure free conditions, i.e., effective stress intensity range, with the crack fully open over the entire load cycle.

σ^* . The study concludes with a discussion of results obtained and their potential extension. As the phenomenon of interest dominates sub-Paris Regime growth rates, the study impacts engineering applications related to high-cycle fatigue and long residual crack growth life.

CONCEPT OF CHARACTERISTIC NEAR-TIP STRESS

For a given distribution of active species, crack-tip diffusion kinetics will be moderated by the local elastic strain field, whose intensity may be represented by σ^* . Applied stress intensity, K , cannot serve this purpose because it ignores local inelastic response, which by definition will be cycle-sequence sensitive.

The Neuber equation relating local elastic stress, $K_t \cdot S$, to local *inelastic* stress, σ , strain, ε , and Modulus, E [20]:

$$(1) \quad K_t S = \sqrt{\sigma \cdot \varepsilon \cdot E}$$

may be extended to inelastic stress at distance, r , from the fatigue crack tip:

$$(2) \quad K / \sqrt{2\pi \cdot r} = \sqrt{\sigma \cdot \varepsilon \cdot E}$$

We assume σ , at a certain characteristic distance from the crack tip, r^* , to represent the magnitude of the instantaneous near-tip stress field. r^* may be treated as a material constant, determined on the premise that at distance, r^* , from the crack tip, application of critical stress intensity, K_c , will induce fracture stress σ_f and fracture strain, ε_f . This limit condition eliminates the possibility of unrealistic computed values of σ under any loading condition. r^* is derived from Eq 2 as:

$$(3) \quad r^* = \frac{1}{2\pi} \left(\frac{K_c^2}{\sigma_f \varepsilon_f E} \right)$$

Equation 3 follows from Eq 2 on the assumption of plane stress conditions at depth r^* from free crack-tip surface. r^* for the material under study was estimated to be about $35 \mu\text{m}$, (0.0345 mm). It may be noted that σ^* at r^* represents a simplistic approximation of actual stress state and gradient that governs crack-tip surface diffusion kinetics. While it is entirely possible that future research will come up with a vastly improved representation of these parameters, the very nature of the relationship is unlikely to change given the physics of the process and its effect on ΔK_{th} .

STRESS INTENSITY AND CHARACTERISTIC NEAR-TIP STRESS

Load interaction effects are accounted for in notch fatigue analysis by simulating cycle sequence sensitive notch root mean stress [21] that in turn will affect computed cumulative damage over a given load cycle. The LSS approach involves solving the Neuber and Ramberg–Osgood equations after accounting for material memory using the Masing model [21,22]. Considering that the crack-tip is a notch and on the assumption that at r^* , it will exhibit the same local monotonic and cyclic inelastic response as a notch, we proceed to extend the LSS approach to the fatigue crack tip. Thus, for monotonic loading, K can be related from Eq 2 through the Ramberg–Osgood equation to σ^* , given modulus, E , strength coefficient, K' , and strain hardening exponent, n' :

$$(4) \quad K = \sqrt{2\pi \cdot r^* \cdot E \cdot \sigma \left[\sigma/E + (\sigma/K')^{1/n'} \right]}$$

For reversed loading:

$$(5) \quad \Delta K = \sqrt{2\pi \cdot r^* \cdot E \cdot \Delta \sigma \left[\Delta \sigma/E + 2(\Delta \sigma/2K')^{1/n'} \right]}$$

The material constants assumed for analysis were selected to reflect mechanical properties of 2024-T3: $E = 73.1$ GPa, $\sigma_f = 469$ MPa, $\varepsilon_f = 0.2$, $n' = 0.13$, $K' = 736$ MPa, $K_c = 37$ MPa $\sqrt{\text{m}}$. Equations 4 and 5, computed using these constants, are graphically described in Fig. 1. They reflect decreased stress concentration that one associates with inelastic notch root response. As local monotonic and reversed stress response are directly related to stress intensity by Eqs 4 and 5, the curves in Fig. 1 serve as transfer functions to estimate local stress as a function of applied stress intensity variation.

ESTIMATING CHARACTERISTIC STRESS UNDER VARIABLE AMPLITUDE LOADING

In notch fatigue analysis using the LSS approach, so called “availability constants” for each linearised segment of the stress–strain curve permit tracking of exhausted stress–strain response under a given arbitrary load sequence in order to faithfully emulate material “stress–strain memory” [22]. In much the same manner, Eqs 4 and 5 can also be transformed into linear segments for cycle-by-cycle numerical simulation of σ^* under a given arbitrary applied load sequence. However, fatigue crack growth analysis involves a moving crack-tip that demands the simultaneous tracking of material stress–strain memory across a string of elements representing the projected crack path. Such elaborate simulation is not required if we assume a stationary crack, a justifiable assumption given near-zero crack extension between periodic overloads under threshold conditions given the loading scheme used in this study, whose description is forthcoming.

FIG. 1

Monotonic and reversed stress excursion, σ , versus stress intensity excursion, K . Curves computed from Eqs 4 and 5 for 2024-T3 Al-alloy.

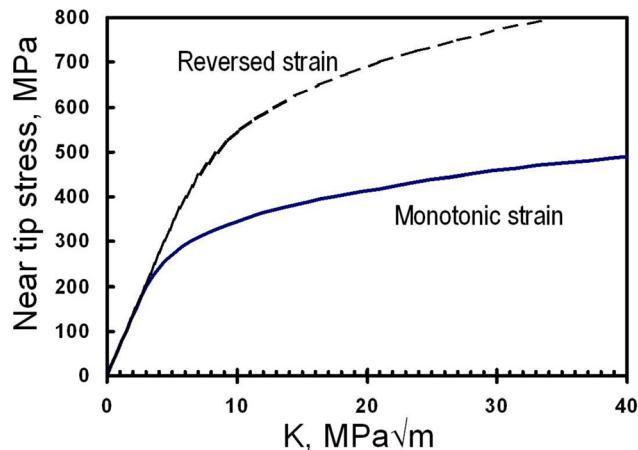


Figure 2(a) shows the general case of an overload cycle superposed on baseline constant amplitude loading. The tensile overload can be preceded or followed by $K_{\min,ov}$ equal to or lower than baseline $K_{\min}$, and be of magnitude, $K_{\max,ov}$ that is equal to or greater than baseline $K_{\max}$. For such a general case, irrespective of $K_{\min}$ immediately preceding the tensile overload, characteristic stress $\sigma_{\max,ov}$ at $K_{\max,ov}$ will be determined by Eq 4 because at $K_{\max,ov}$ material response may be assumed to follow the monotonic stress-strain curve:

$$(6) \quad K_{\max,ov} = \sqrt{2\pi \cdot r^* E \cdot \sigma_{\max,ov} \left[\sigma_{\max,ov} / E + (\sigma_{\max,ov} / K')^{1/n'} \right]}$$

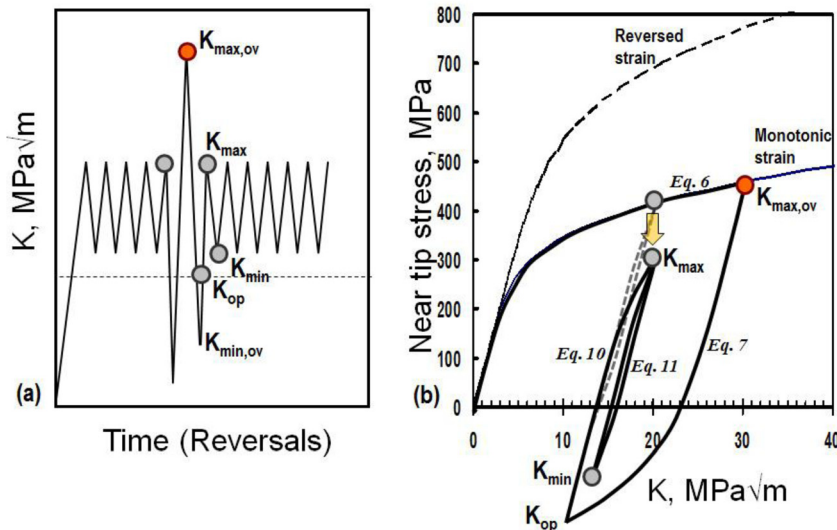
The local stress excursion on the falling load half-cycle following the tensile overload will be determined by the downward stress intensity excursion to $K_{\min,ov}$. This excursion is given by the difference, ΔK_{down} , between $K_{\max,ov}$ and the higher of the two values, K_{op} and $K_{\min,ov}$. Assuming K_{op} is positive, this difference cannot exceed $K_{\max,ov}$. Therefore, crack-tip response will be governed by reversed stress-strain response per Eq. 5:

$$(7) \quad \Delta K_{\text{down},ov} = \sqrt{2\pi \cdot r^* E \cdot \Delta \sigma_{\text{down},ov} \left[\Delta \sigma_{\text{down},ov} / E + 2(\Delta \sigma_{\text{down},ov} / 2K')^{1/n'} \right]}$$

where

$$(8) \quad \Delta K_{\text{down},ov} = K_{\max,ov} - \max(K_{op}, K_{\min,ov})$$

FIG. 2 (a) Generalised case of overload applied over baseline cycling. The schematic shows $K_{\min,ov}$ following $K_{\max,ov}$ but the reverse is also possible. The crack tip will not see the excursion below K_{op} . (b) Translation of sequence in (a) into near-tip stress response. Applicable equation is indicated against individual segments. Arrow indicates drop in near-tip residual stress with respect to the one before the tensile overload. This drop will induce increased ΔK_{th} .



The rising excursion, ΔK_{up} from $K_{\text{min,ov}}$ to baseline K_{max} cannot exceed the preceding downward excursion from $K_{\text{max,ov}}$. Therefore, Eq 5 for fully reversed stress response will again apply:

$$(9) \quad \Delta K_{\text{up}} = K_{\text{max}} - \max(K_{\text{op}}, K_{\text{min,ov}})$$

$$(10) \quad \Delta K_{\text{up}} = \sqrt{2\pi \cdot r^* E \cdot \Delta \sigma_{\text{up}} \left[\Delta \sigma_{\text{up}} / E + 2 \left(\Delta \sigma_{\text{up}} / 2K' \right)^{1/n'} \right]}$$

Finally, near-tip response to the baseline falling load half-cycle leading to governing unloaded condition determining ΔK_{th} is given by

$$(11) \quad \Delta K_{\text{down}} = \sqrt{2\pi \cdot r^* E \cdot \Delta \sigma_{\text{down}} \left[\Delta \sigma_{\text{down}} / E + 2 \left(\Delta \sigma_{\text{down}} / 2K' \right)^{1/n'} \right]}$$

Finally, the falling load half-cycle of baseline cycling immediately following the overload will determine characteristic near-tip residual stress, σ^* :

$$(12) \quad \sigma^* = \sigma_{\text{max,ov}} - \Delta \sigma_{\text{down,ov}} + \Delta \sigma_{\text{up}} - \Delta \sigma_{\text{down}}$$

We proceed on the premise that Eq 12 determines σ^* (and therefore, ΔK_{th}) over the baseline cycling following the overload. **Figure 2(b)** illustrates Eq 12 by showing K -stress excursions associated with the concerned sequence of load excursions involving overload.

It is pertinent and important to underscore the significant role of the crack closure phenomenon as reflected in Eqs 8 and 9 in truncating near-tip response to applied loads falling below closure load. Such truncation differentiates near-tip response from notch root response. As seen above, it simplifies analysis of near-tip response thanks to the consideration that the four stress excursions involved in the application of an overload cycle that affect σ^* will always be in receding order of magnitude, independent of the magnitude of the overload and underload as well as their sequence! If successive reversals in local stress were to be in increasing order, numerical simulation would be inevitable⁷, rather than the closed form expressions used above.

It is also pertinent to note that σ at $K_{\text{max,ov}}$ is uniquely determined by Eq 4, and may therefore be history independent. This may explain why compressive overloads preceding tensile overloads do not significantly impact subsequent retardation, except, perhaps, through closure.

Finally, judging from Eq 12, σ^* under baseline cycling after an overload is controlled by the magnitude of four consecutive stress intensity reversals involved in the overload/underload sequence and has less to do with the *percentage* overload involved. Practically all conventional data correlate crack growth retardation with percentage tension/compression overload applied. This may be a flawed approach considering the possibility that the same percentage overload will induce different

⁷Material stress-strain response exhibits memory effect as accounted for by numerical modeling by LSS approach. In case of successive local stress reversals of increasing magnitude, material response within a reversal involving a switch from Eq 5 to Eq 4 cannot be ruled out.

magnitude of downward near-tip stress excursions depending on baseline $K_{\max}$. These play a crucial role in determining σ^* after an overload.

Experimental Procedure to Determine ΔK_{th}

Equation 12 represents the general equation for σ^* under baseline $K_{\max}$ following an overload. For the partial case of constant amplitude cycling, σ^* can be readily estimated from Eq 4. As Eqs 6–12 will apply to any combination of tensile and compressive overloads, irrespective of their magnitude, strictly controlled and reproducible σ^* will apply under subsequent baseline cycling. It would follow that crack growth under baseline cycling following overloads must be governed by highly reproducible values of ΔK_{th} tied to σ^* . This forms the basis for the proposed experimental procedure to establish the relationship between ΔK_{th} and σ^* .

GENESIS OF STANDARD LOAD SEQUENCE TO DETERMINE ΔK_{TH}

From the standpoint of engineering application it is pointless to determine ΔK_{th} under cycling involving partial crack closure. Even a small percentage error in estimate of closure will grossly distort $\Delta K_{th,eff}$ because in absolute terms, the magnitude of K_{op} can far exceed that of $\Delta K_{th,eff}$. The focus of this study is therefore on estimating ΔK_{th} in the assured absence of crack closure. To preclude plasticity-induced closure, percentage magnitude of tensile overload is restricted, baseline cycling is at high stress ratio and where necessary, compressive loads are applied⁸. Unless otherwise qualified, it may be assumed that ΔK refers to ΔK_{eff} and that ΔK_{th} refers to $\Delta K_{th,eff}$.

ΔK_{th} is determined by “no growth conditions,” defined as no discernible crack extension over half a million cycles. Based on the actual reproducibility of compliance based crack size estimates with the compact tension C(T) specimen used in this study, discernible crack extension is assumed to be 0.025 mm, implying ΔK_{th} applies to a crack growth rate less than 0.5×10^{-7} mm/cycle. This will necessarily be a conservative estimate because crack extension during the overloads cannot be subtracted, except through quantitative fractography. One may expect that at threshold, crack extension during overloads may be restricted to the Paris Regime of 10^{-4} to 10^{-2} mm/cycle. Greater overload crack extension would render the ΔK_{th} estimate even more conservative. In the event overload crack extension exceeds 0.025 mm, “no growth” condition cannot be satisfied and there will be no ΔK_{th} estimate.

Conventional laboratory testing practice per ASTM E647 revolves around testing under controlled stress ratio. Applied stress ratio is of little practical relevance under random load time histories because computed crack extension is based on maximum load and effective load range over a given load cycle. Based on the same consideration, the proposed load sequence and testing practice associates ΔK_{th} with a given $K_{\max}$ rather than R . Therefore, testing is performed under $P_{\max} = \text{Const.}$ and ΔK is reduced by increasing $P_{\min}$. This permits characterization of ΔK_{th} under increasing $K_{\max}$. As $K_{\max}$ progressively increases with crack size each ΔK_{th} estimate will pertain to a different $K_{\max}$.

⁸As shown earlier, compressive overloads preceding tensile overloads do not impact σ^* under subsequent cycling.

The proposed technique has little in common with the one proposed by Marci [23] and others to characterize ΔK_{th} at $K_{max} = \text{Const}$. Firstly, $K_{max} = \text{Const}$ is not the same as $P_{max} = \text{Const}$ because the former still involves load shedding induced history effects that render the result rate sensitive. The proposed technique does not involve load shedding. Reduction in ΔK is achieved exclusively through *increase* of P_{min} , leaving absolutely no history induced distortion of either monotonic plastic zone size, or, of maximum stress in the cyclic plastic zone. Secondly, the proposed technique sets the goal of inducing *controlled variations* in near-tip residual stress through application of programmed overload-underload combinations. One of the most important goals of conventional practice is the exact opposite—of excluding as far as possible, any history induced near tip residual stress that may render ΔK_{th} irreproducible! Thirdly, the proposed technique treats applied stress ratio as irrelevant to ΔK_{th} on the premise that this is indeed the case for a fully open crack. This permits crack size measurements through slow cycling under large applied load range; that is not possible by prevailing standard practice, leading to potential error in compliance based crack size estimates because of reduced data count as ΔK tends to ΔK_{th} . Most importantly, the goal of the proposed technique is to characterize ΔK_{th} as a unique function of σ^* , rather than attempt to relate it to LEFM parameters, such as K_{max} or stress ratio.

It is important to note that $P_{max} = \text{Const}$. ensures compatibility with conditions of engineering application that are far removed from K -control. Furthermore, as ΔK reduction is through increasing P_{min} , there is practically no restriction on the rate of load shedding. As will be seen below, this permits generation of several ΔK_{th} estimates from a single specimen, with each pertaining to a different K_{max} and also, a different σ^* , thereby permitting generation of more non-duplicative data points from fewer test coupons.

Only two parameters are varied in the course of the experiment leading to determination of ΔK_{th} . These are P_{min} and baseline step size. Baseline cycling commences at a minimum stress ratio, R , associated with a fully open crack, typically, $R = 0.4$. Cycling involves application of a step of baseline cycling followed by a slow overload/underload sequence during which crack size is estimated from unloading compliance. As ΔK -reduction is through increase in P_{min} , and given the significant increase in stress ratio before onset of no growth conditions, initial stress ratio is not a crucial parameter.

Three possibilities emerge at the conclusion of each block of loading. In the event crack extension is detected, baseline P_{min} is incremented in order to reduce ΔK by 5 % and baseline step size is reset to 10 000 cycles. If no crack extension is detected, baseline step size is doubled over the next block. Enforcement of these two conditions leads to eventual crack arrest (no crack extension after 500 000 cycles), which is the third possibility. Baseline ΔK at crack arrest, incremented by 2.5 % (average of two last ΔK applied) is treated as ΔK_{th} deemed to have been determined with better than 5 % resolution. At this point, ΔK is reset to higher magnitude associated with $R = 0.4$, baseline step size restored to 10 000 cycles and the process continued to the next crack arrest. Obviously, the next estimate will correspond to a higher value of K_{max} . At some point, K_{max} will have risen to a magnitude either leading to specimen failure, or, non-satisfaction of “no growth” criterion because of sizeable crack extension during the overload. At this point, testing is terminated. The entire test process is automated, with each specimen yielding a string of ΔK_{th} values.

Figure 3 summarises the loading scheme to determine ΔK_{th} . The desire to select an initial stress ratio such that the crack is likely to be largely open is driven by consideration of minimum desired crack increment before the onset of threshold. Thus, if $R = 0.1$ were to be chosen, increments in P_{min} involving partial crack closure would “be wasted,” having little consequence to retarded growth rate until the crack is fully open at about $R = 0.4$ or greater. Note that as $P_{max} = \text{Const}$, the rise in monotonic plastic zone size will be gradual as is the case in engineering practice, leaving no room for uncontrolled load interaction effects. Moreover, incremental P_{min} between steps cannot influence σ^* . Furthermore, as unloading compliance is measured from the slow and large overload cycle, extremely consistent and reproducible measurements of crack size are possible using conventional crack opening displacement (COD) transducers specified by ASTM E647, right down to threshold conditions. This is in contrast to the overheads associated with prevailing standard practice, whereby the quality of crack size estimates progressively deteriorates with decreasing ΔK . Finally, baseline cycling with step size varying between 10 000 and 500 000 cycles can be performed at as high a frequency as is possible, of course, with due allowance to satisfy compliance of surface diffusion kinetics with those representative of service usage conditions. This may negate the use of ultra-high frequencies as applied in giga-cycle fatigue.

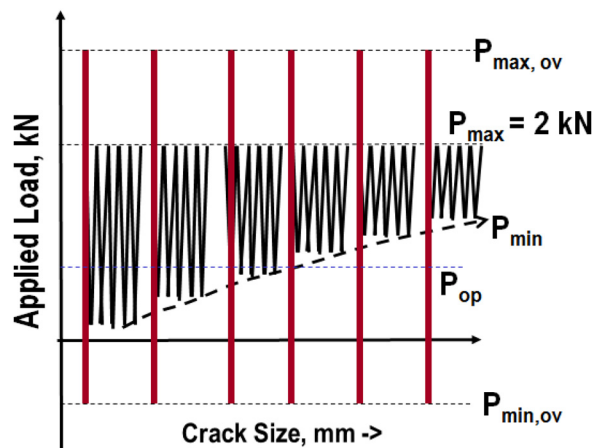
TEST SPECIMENS AND EQUIPMENT

All tests were performed on 50 mm wide, 5 mm thick C(T) specimens cut from 2024-T3 Al-alloy sheet material in both LT and TL directions. Knife-edge groove to mount the COD gauge and a crack initiator to $a/W = 0.2$ were CNC wire cut to ensure reliable COD gauge mounting and reproducible pre-cracking and compliance measurements.

Backlash free load train arrangement was implemented by the insertion of tight fitting roller bearings into the clevis and reduced diameter holes with tight fit pins on the specimen. Backlash free gripping is required to enable the application of compressive overloads. However, inclusion of compressive loads as a test condition in the present study did not significantly affect observed ΔK_{th} response. The desired

FIG. 3

General description of ΔK shedding scheme under constant P_{max} to determine ΔK_{th} .



effect was achieved by taking $P_{\min}$ below closure load. This suggests that compressive loads can be avoided altogether and that conventional clevis grips and specimen hole diameters as per ASTM E647 may suffice if the objective is to establish the relationship between ΔK_{th} and σ^* .

Testing was performed on a 5 kN servo-hydraulic test system equipped with a BiSS 2370SS digital controller. The controller firmware provides for up to 32 kHz servo-loop update frequency with adaptive peak/valley control to enforce accurate load cycling at high frequency. The servo-actuator was equipped with a Parker voice coil servo valve permitting cycling at up to 250 Hz. Baseline cycling was performed at 150 Hz after confirming that peak/valley loading error at this frequency did not exceed 4 N over the load range capacity of the machine and over crack growth interval $0.2 < a/W < 0.8$. This permitted attaining a threshold measurement by “no growth” every 3–5 h.

With the exception of a single test performed at $P_{\max} = 1$ kN, all other testing was performed under conditions of constant baseline maximum load, $P_{\max} = 2$ kN. The results from the test at $P_{\max} = 1$ kN are indistinguishable from the rest of the tests when viewed in terms of stress intensity, confirming that near-tip residual stress is controlled by stress intensity factor with secondary variables such as applied load playing an insignificant role.

Precracking is performed to grow a uniform fatigue crack front that is free from the potential influence of the crack initiator notch. At the conclusion of precracking, ΔK is assumed to be significantly greater than ΔK_{th} . To avoid any sequence effects, precracking is also performed at the same $P_{\max}$ as during subsequent cycling to determine ΔK_{th} but with stress ratio $R = 0.1$ to a crack size of 14 mm (0.28 W). The process of cycling during precracking allows compliance measurements to settle down to highly reproducible values within 0.0005 W at the commencement of threshold estimates. The resolution and reproducibility of compliance measurements essentially determine both capability and quality of this experimental procedure.

The application software developed for this study provides for determination of baseline constant amplitude ΔK_{th} values and for the following specifications of tension/compression overload application: magnitude and number of overload cycle including maximum, minimum and cycle frequency, sequence (maximum point preceding or following minimum), and, optional hold time (dwell) at maximum or minimum prior to resumption of baseline cycling. These appear to cover most cases of simple variable-amplitude loading referred to in the literature.

Exploratory test results showed that for the material in question, the number of overloads and dwell time do not appear to noticeably influence ΔK_{th} . Literature data, however, do suggest that these can affect load interaction. Multiple overloads cause crack extension into the overload plastic zone, leading to increased crack closure at the resumption of baseline cycling. Cyclic stress-strain response in a qualitative sense can differ significantly from material to material. Thus, some materials will exhibit cyclic strain hardening, while others will see strain softening. In this case, number of overloads will play a role. Some materials such as Titanium alloys will exhibit rate sensitivity and may therefore be sensitive to dwell time. Therefore, the additional software features of programmability of number of overloads and dwell time may be essential to characterize threshold response of other materials that are likely to exhibit sensitivity to such features.

In our experiments, overload levels of 30, 50, 62.5, 75, and 100 % were used. As indicated by forthcoming discussion, application of overloads in excess of 62.5 % involved partial crack closure at threshold as indicated by highly inflated values of ΔK_{th} . These data were discarded as invalid.

Test Results and Discussion

Figure 4 shows vast variation in closure-free ΔK_{th} depending on the specific load sequence applied and its parameters. Under constant amplitude loading, there is a small, but well defined *decrease* in ΔK_{th} with increasing K_{max} , of the order of 1 % for every 1 MPa $\sqrt{m}$. Application of tensile overloads causes a steady *increase* in ΔK_{th} with K_{max} up to values as high as five times by comparison to ΔK_{th} under constant amplitude loading. Such increase progresses more rapidly with increasing percentage magnitude of the tensile overload. The increasing trend as well as the rate of increase of ΔK_{th} does not appear to be affected by the application of compressive loads preceding the tensile overload, irrespective of the magnitude of such compressive loads. However, ΔK_{th} is extremely sensitive to P_{min} immediately following the tensile overload in the overload cycle. Thus, a compressive overload following a tensile overload can reduce ΔK_{th} to constant amplitude level. This can be seen from the three sets of data from TL coupons with 50 % tensile overload followed by P_{min} leading to overload stress ratio, $R = 0.1, 0.3$, and 0.5 . Their trends are indicated by the three arrows in the figure. When P_{min} following the overload drops down to just 50 % of overload P_{max} , ΔK_{th} steadily increases with K_{max} . However, when it drops down to 30 %, ΔK_{th} actually begins to decrease with K_{max} , whereas when the specimen is practically just unloaded, with P_{min} dropping to 10 % of overload P_{max} , ΔK_{th} response is practically the same as under constant amplitude loading. Classical understanding of fracture mechanics cannot explain all these results, let alone unify them. The wide dispersion of ΔK_{th} data when plotted against K_{max} appear to support this conclusion.

Figure 5 shows ΔK_{th} values from **Fig. 4** re-plotted against σ^* for individual loading conditions. They fall into a narrow scatter band that appears to confirm a close relationship between the two parameters. Three distinct regions may be identified in

FIG. 4

Closure free ΔK_{th} versus baseline K_{max} . Baseline data on specimens with TL orientation were obtained with high compressive load applied during slow cycles. Suffix "Min" indicates overload stress ratio with $K_{min,ov}$ following $K_{max,ov}$. All other tests with overloads involved return to baseline K_{min} following stated percentage tensile overload.

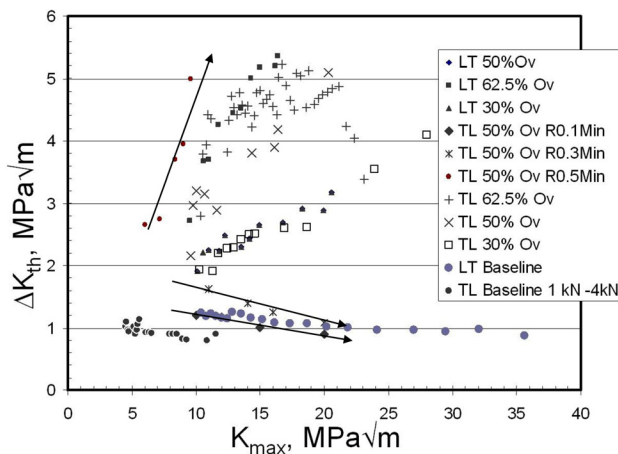
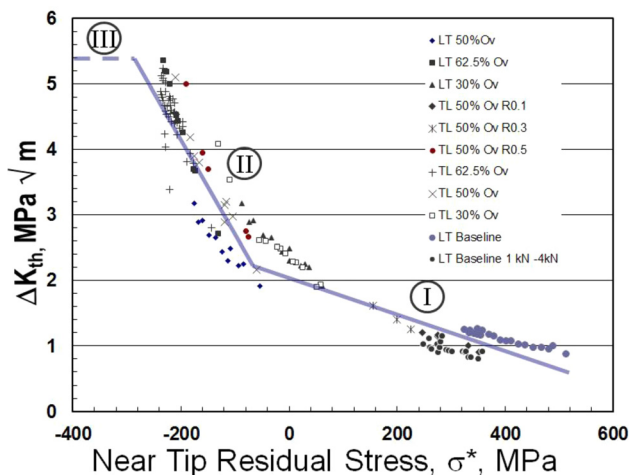


Fig. 5 using the rationale behind the BMF model. Region I pertains to typical conditions of constant amplitude loading, associated with high enough crack-tip stresses to induce saturated surface diffusion within the time period associated with rising load half-cycle. Further increase in crack-tip stress beyond a point causes only a marginal, though visible decrease in ΔK_{th} . Region II is the transient region of diffusion kinetics with its dramatic effect on ΔK_{th} . Over this region, ΔK_{th} will be extremely sensitive to variation in σ^* . One may expect that the nature of diffusion kinetics will force this region to shift to the left or to the right, depending on test frequency, partial pressure of moisture and temperature. Region III is the region of diffusion kinetics diminishing to vanishing proportions as in high vacuum. One may conclude that at near-tip residual stress levels associated with this region, the fatigue crack tip essentially behaves as it would under conditions of high vacuum. Hence, the “plateau” in ΔK_{th} , associated with total absence of BMF as in high vacuum. This condition may manifest in atmospheric fatigue through inhibition of surface diffusion by reduced local stress from prior tensile overload, reduced partial pressure of active species, or, very short rise time in load as at very high cycling frequency. The presence of Region III as a plateau may appear speculative given the absence of associated experimental data points in **Fig. 5**. One may note that upon the onset of considerable yield in compression, compressive stresses will tend to be less sensitive to further drop in applied load. The plateau in threshold response associated with more severe but closure free tensile overloads has been noted in earlier work [6] and also appears when data obtained in this study are presented in a different format (see **Fig. 7** below). Importantly, representation of Region III as a plateau underscores the significance of high vacuum as a limiting factor, whereby ΔK_{th} cannot increase beyond a certain level. The shape of this band is similar to well-known fatigue crack growth rate response when plotted against parameters describing test environment and attributable to crack-tip diffusion kinetics of active species. For example, a similar shape of curve would be obtained if crack growth rate at controlled ΔK is plotted against partial pressure of moisture, or, against cycling frequency at given partial pressure of moisture [19,24,25].

FIG. 5

Closure free ΔK_{th} estimates from **Fig. 4** plotted against near-tip residual stress, σ^* , computed as stress at distance, r^* from the crack tip at the commencement of threshold fatigue load cycle immediately following overload sequence. All 160 data points appear to fall into a narrow scatter band over two linear segments. The third segment, (III), has been added to suggest the expected limiting nature of high vacuum threshold response.



The reasonable correlation of ΔK_{th} with σ^* , irrespective of the type and magnitude of overloads raises the specific question of what makes the trend line change direction depending on the magnitude of underload following the tensile overload. **Figure 6** shows the results of near-tip stress simulation for the three cases marked by arrows in **Fig. 4** for 50 % tensile overload followed by underload to three different ratios of $R_{ov} = 0.1, 0.3$, and 0.5 . **Figure 6(a)** shows near-tip stress response to the three cases at baseline $K_{max} = 10 \text{ MPa}\sqrt{\text{m}}$. In order to render data in a comparable manner, the simulation was concluded at equal extent of unloading from baseline K_{max} as indicated in the figure. Furthermore, crack closure was ignored in computing response, even for the case of $R_{ov} = 0.1$ (one may speculate that immediately after a tensile overload, the crack remains fully open). **Figure 6(a)** illustrates how inelastic unloading following a tensile overload can diminish its effect of retardation due to increased level of σ^* . **Figure 6(b)** shows computed near-tip stress for the three cases plotted as a function of baseline stress intensity, K_{max} . Interestingly, up to a certain baseline K_{max} , the trend lines for $R_{ov} = 0.1$ and 0.3 are in the opposite direction from that for $R_{ov} = 0.5$, just as one would expect from the trend lines marking empirical results in **Fig. 4**. One may note that the data in **Fig. 4** for $R_{ov} = 0.5$ are within the interval of baseline K_{max} up to $12 \text{ MPa}\sqrt{\text{m}}$, where σ^* reduces with increasing K_{max} in contrast to the other two stress ratios where it shows an increasing trend. This appears to reinforce the possibility that indeed, ΔK_{th} may be closely related to sequence-sensitive near-tip stress, σ^* . **Figure 6** certainly appears to explain the deleterious effect of underloads following overloads through their effect on σ^* that in turn affects ΔK_{th} .

FIG. 6 Simulated near-tip stress response to varying degree of underload following 50 % tensile overload. (a) Near-tip Stress-K response to a 50 % tensile overload at baseline $K_{max} = 10 \text{ MPa}\sqrt{\text{m}}$ followed by unloading to 0.1, 0.3, and 0.5 of tensile overload, re-loading to baseline K_{max} concluded by $2 \text{ MPa}\sqrt{\text{m}}$ unloading. (b) Near-tip stress computed for the same percentage tensile overload and cases of subsequent underload ratio followed by at same level of unloading ($2 \text{ MPa}\sqrt{\text{m}}$) plotted as function of baseline K_{max} .

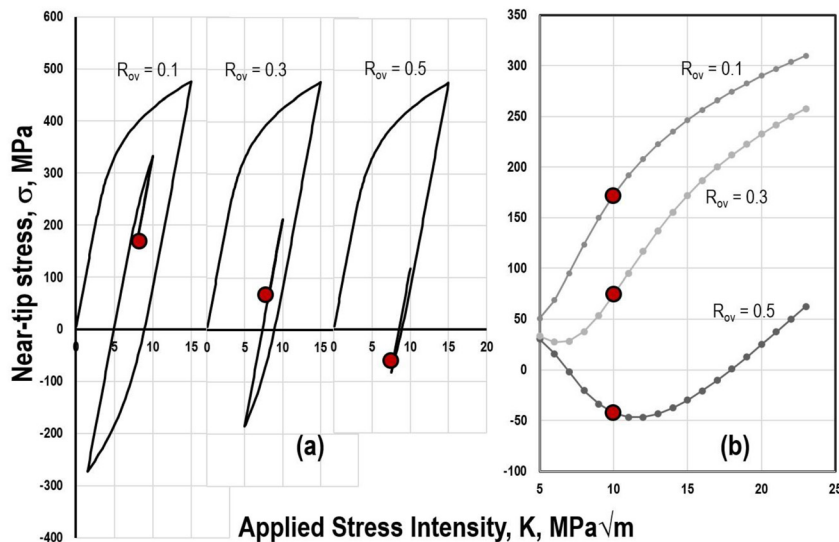
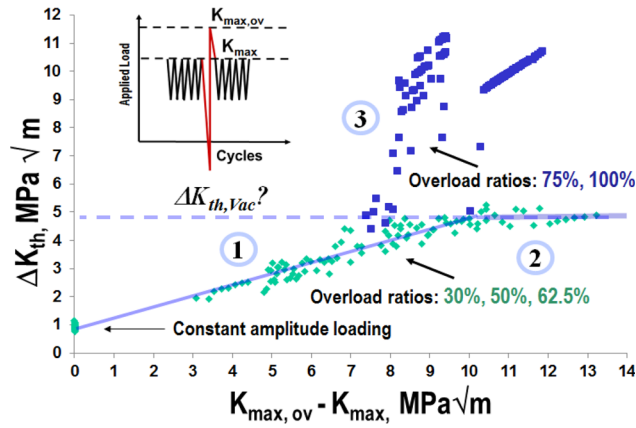


FIG. 7

ΔK_{th} estimates plotted against drop in max stress intensity from tensile overload to baseline loading. Data include measurements obtained under 75 and 100 % overloads that induce partial crack closure. Data obtained in experiments involving tensile overload followed by compressive loads excluded from the plot. Closure free estimates appear to fall into narrow scatter band over two linear segments. Plateau 2 is suggestive of response saturation.



Assuming elastic unloading over twice yield stress, a drop in K from $K_{max,ov}$ will induce a proportional drop in near-tip stress. One can then recast data in **Fig. 5** for the partial case of tensile overloads against the drop in K associated with transition from overload $K_{max,ov}$ to baseline K_{max} . These appear in **Fig. 7**. With the exception of the 75 and 100 % overload data that have been included to indicate how partial crack closure can distort ΔK_{th} , irrespective of the magnitude of applied baseline K_{max} , ΔK_{th} values fall within a narrow scatter band when plotted against a simple computed parameter given as the difference between overload and baseline K_{max} . The characteristic linear relationship on the rising part of the curve (Zone 1) appears to confirm the assumption of elastic near-tip residual stress response upon unloading as a function of K . The plateau (Zone 2) underscores attainment of “high vacuum” like conditions. The data in Zone 3 are from tests under 75 and 100 % overloads that induced partial crack closure. K_{op} is linearly proportional to $K_{max,ov}$. Beyond a certain percentage overload, there is the threat of $K_{max} - K_{op}$ falling below ΔK_{th} . This would invalidate ΔK_{th} estimate given a partially closed crack.

Available data for 2024-T3 including those in **Fig. 7** permit quantification of criteria to disqualify ΔK_{th} estimates as being invalid due to partial crack closure. This may be quantified from two equations, the approximating linear equation for ΔK_{th} as a function of $K_{max,ov} - K_{max}$ from **Fig. 7** given as

$$(13) \quad \Delta K_{th} = \Delta K_{th0} + 0.4K_{max}(k_{ov} - 1), \quad \text{while} \quad K_{max,ov} - K_{max} \leq 10 \text{ MPa}\sqrt{m}$$

and from the equation for ΔK_{eff} given as

$$(14) \quad \Delta K_{eff} = K_{max}(1 - k_{op}k_{ov})$$

where $\Delta K_{th0} = 1 \text{ MPa}\sqrt{m}$, $k_{op} = K_{op}/K_{max}$ and $k_{ov} = K_{max,ov}/K_{max}$.

Equating Eqs 13 and 14 as a condition for fatigue crack arrest, we obtain

$$(15) \quad k_{ov} = \frac{1.4 - \Delta K_{th0}/K_{max}}{0.4 + k_{op}}$$

Assuming that at $R = 0.1$, the onset of crack closure at surface layers will occur at about 40 % of maximum load (in this case, the overload), a 75 % or more tensile overload is likely to arrest the fatigue crack due to partial crack closure at threshold. This may explain the highly exaggerated values of ΔK_{th} obtained in experiments with 75 and 100 % overload. In future experiments, Eq 15 can be used to determine maximum percentage overload that can be used to obtain valid estimates of ΔK_{th} pertaining to a fully open crack.

An important implication of the results in Figs. 5 and 7 is that a fully open fatigue crack can arrest immediately upon application of a tensile overload, even if subsequent cycling proceeds at up to five times the magnitude of constant amplitude ΔK_{th} . The controlling criterion is the drop in stress intensity from that at tensile overload.

CRACK TIP RESPONSE, CRACK CLOSURE AND COMPRESSIVE LOADS

As shown by Eqs 4–12, post overload σ^* is insensitive to downward load excursions preceding a tensile overload. However, the equations indicate that downward local inelastic stress excursion following a tensile overload will induce increased σ^* during subsequent baseline cycling, practically eliminating any beneficial effect of the tensile overload. These analytical conclusions are matched by actual ΔK_{th} that lie in the same group as constant amplitude data in Fig. 5.

As is generally accepted, crack closure will see an immediate drop upon application of a compressive load. This is because the crack wake will yield in compression and it will take some crack extension into the overload plastic zone for closure to recover, and then rise to a level associated with the overload. Post overload crack closure develops as a consequence of new (stretched) wake evolving with crack extension into the overload plastic zone. Crack extension during the overload may be assumed to be negligible by comparison to wake depth contributing to closure, or for that matter, the monotonic plastic zone size. Therefore, one may conclude that neither the sequence nor the magnitude of compressive overload should seriously affect post overload closure evolution provided the residual tensile deformation within the overload plastic zone remains unaffected.

It emerges that the crack closure phenomenon cannot explain the dramatic sequence effect of compressive load on post overload retardation. Compressive overload applied after a tensile overload restores ΔK_{th} to constant amplitude levels. Even so, if baseline cycling proceeds at a low enough stress ratio to be affected by changes in crack closure, delayed retardation will be observed given crack increment required for increased closure to evolve. However, if baseline cycling is at high stress ratio unaffected by closure, there will be practically no retardation. Furthermore, if baseline ΔK is into the Paris Regime and “beyond the influence of ΔK_{th} ,” delayed retardation is inevitable because over this range of growth rates crack extension by slip overshadows BMF in scale.

The increasing nature of ΔK_{th} with K_{max} under tensile overloads will throw up crack growth rate behaviour that cannot be explained by crack closure. As shown by Fig. 5 as well as Fig. 7, under periodic tensile overload of a fixed magnitude, ΔK_{th} will progressively increase to saturation. The rate of this increase is determined by the percentage of overload magnitude. It would follow that at a particular percentage overload, the rate of this increase can exceed the rate of increase of ΔK due to crack extension. This will lead to gradual decrease in growth rate and potential crack arrest. Such response cannot be explained by conventional wisdom that assumes

constant ΔK_{th} . If $\Delta K_{th} = \text{Const}$, once a crack commences growth under periodic overloads, it cannot retard, let alone arrest.

The above observations underscore the futility of interpreting variable-amplitude fatigue response through closure alone. Synergy of two independent mechanisms, crack closure, and sequence sensitive ΔK_{th} explains contrasting findings in the literature regarding the effect of tensile and compressive overloads on fatigue crack growth. By focusing on the monotonic plastic zone, the Wheeler and Willenborg models overlook the important role of cyclic hysteretic stress response within the cyclic plastic zone.

Given the results of this study, the success of crack growth estimates using the Wheeler, Willenborg, and Elber models comes as a surprise. Their success may be restricted to the Paris Regime where cycle-sequence sensitive response may be negligible. It may also be possible that fortuitous selection of model constants and suitable adjustment of closure levels can compensate for errors induced by the erroneous assumption of $\Delta K_{th} = \text{Const}$.

Lack of reliable techniques for closure measurement, particularly under near threshold growth rates appear to have led to questions regarding the very credibility of the phenomenon [26]. Ironically, while near tip stress has no bearing on crack closure, near-tip inelastic strain response may arguably be the most compelling indicator of actual crack opening load. This was confirmed by highly localized measurements of near-tip strain using laser interferometry. They illustrate that *inelastic* near-tip strain response and consequent hysteresis seen in applied load versus local strain response are a direct indication that the crack is open [25]. Crack closure is an extremely localized phenomenon. Far field measurements such as crack mouth opening displacement (CMOD) or Back Face Strain (BFS) are incapable of characterizing near tip hysteretic stress-strain response. ASTM E647 practice for closure measurement is built around detecting compliance change from CMOD or BFS. Compliance change reflects wake contact kinetics. However, in doing so, it does not discriminate between far field and near field wake in the manner in which crack closure may be sensitive to them.

The experimental data obtained in the study, backed up by Eq 15, suggest that as long as tensile overload does not exceed 65 % of baseline, closure will be absent at threshold by satisfaction of the criterion $K_{max} - \Delta K_{th} > K_{op}$. Considering that the material under study is known to exhibit reasonably high value of closure, one may expect this to be a conservative criterion for most other materials of engineering interest.

EFFICACY OF TEST PROCEDURE

ΔK_{th} data points obtained in this study represent a vast sample size, that would have been impractical to obtain using prevailing standard practice such as ASTM E647. The reproducibility of data and the consistent trends exhibited under a variety of loading conditions are noteworthy. This should not come as a surprise given the inherent efficacy of the test technique that lies in the combination of high frequency low amplitude cycling to grow the crack, combined with low frequency large amplitude cycling for accurate measurement of crack size from compliance that also ensures highly controlled values of σ^* .

Baseline cycling at 150 Hz ensured establishment of a valid ΔK_{th} data point by “no growth” criteria every 3 to 5 h. State-of-the-art signal processing and digital servo-control techniques combined with affordable voice coil servo-valve technology, driven

by robust algorithms for ΔK control made it possible to completely automate the test process. Even conventional automated test systems can readily deliver 50–80 Hz of low-amplitude cycling associated with threshold testing. The absence of stringent load shedding conditions eases the process of obtaining valid data points. Therefore, the proposed technique can be readily implemented on most conventional test systems.

Quality of ΔK_{th} measurements is determined by two parameters, reproducibility of crack size measurements and percentage decrease (or increase in the case of “finite growth” criterion) in ΔK . Both will determine crack growth interval between ΔK_{th} measurements. The latter will determine the resolution of ΔK_{th} measurement. As the magnitude of the slow load cycles is retained unchanged through the test, the quality of compliance measurements is unaffected by the frequency and load magnitude of baseline cycling. This renders consistency to the quality of crack size measurement and thereby, also to ΔK_{th} estimates. Finally, given their close relationship, reproducible near-tip residual stress imposed by the slow overload cycles obviously also ensures reproducible ΔK_{th} .

ASTM E647 provisions for both $R = \text{Const}$ as well as $K_{max} = \text{Const}$ are less conducive to reproducible crack size measurements given the combined effect of increasing K with crack size and requirement for decreasing ΔK , that lead to a rapidly decreasing Load-COD data window for compliance measurement.

APPLICATION OF TEST RESULTS

The dramatic effect of periodic overloads on ΔK_{th} has been studied in early work [27]. However, at that time, no conclusive explanation for the effect was available, the effect of closure was not isolated from that of residual stress and of course, the data could not be generalized for engineering application. Later work focused on closure-free ΔK_{th} , but at the time the concept of σ^* had not crystallized [6]. A more recent review addresses this subject in considerable detail, but neither a standard procedure for ΔK_{th} estimation nor analytical equations to estimate σ^* had been formulated [28]. The output from this study closes that remaining gap.

Given the universality of the residual stress effect in metal fatigue, one may assume that for every metallic material in an engineering application, there must exist a “Master Curve” defining the relationship between ΔK_{th} and σ^* . The possibility of numerical simulation of σ^* in cycle-by-cycle crack growth analyses for spectrum loading underscores the engineering significance of a standard laboratory practice to determine the master curve as an important addition to conventional testing for crack growth testing per ASTM E647. It appears this curve may be approximated by three linear segments as shown by the dash line in Fig. 5. The top and bottom segments represent extreme conditions of saturated or zero diffusion activity, with a middle transient between the two.

Crack extension over a load cycle may be described by the general relationship:

$$(16) \quad \Delta a = \frac{da}{dN} = C \cdot \Delta K_{eff}^m \frac{[1 - (\Delta K_{th}/\Delta K_{eff})]^p}{[1 - (K_{max}/K_c)]^q}$$

where C , m , p , and q are material constants, with $\Delta K_{eff} > \Delta K_{th}$ and $K_{max} < K_c$. ΔK_{th} is not a material constant, but treated strictly as a cycle-sequence sensitive variable to be re-computed for each load cycle in a given an arbitrary load sequence as a function of

σ^* for that cycle. C and m describe Paris Regime response while p corrects for the influence of ΔK_{th} on sub-Paris Regime response. With ΔK increasing into the Paris Regime, influence of ΔK_{th} will diminish to vanishing proportions and crack growth behavior will tend to that simulated by conventional closure based models.

Equation 16 implies an asymptotic influence of ΔK_{th} on da/dN . However, the process for its determination ties it to a finite, albeit low crack growth rate. Obviously, a correction needs to be introduced to ensure that the approximated near threshold crack growth rate curve will pass through ΔK_{th} . Finally, one may note that “ p ” controls the shape of the near-threshold segment of the da/dN versus ΔK curve only for a pre-determined ΔK_{th} . It may not be appropriate to assume that this “constant” will remain so over the entire range of sequence sensitive ΔK_{th} . In all possibility, an additional relationship needs to be established between “ p ” and ΔK_{th} . The experimental procedure to determine such a relationship is the focus of work in progress.

TRANSIENT ΔK_{TH} CHARACTERIZATION

The test conditions imposed in this study were engineered to ensure tightly controlled and stable σ^* in order to obtain reproducible ΔK_{th} values over a wide range of variation. This was achieved through repeat overloads under practically stationary crack tip conditions. Engineering application of ΔK_{th} demands cycle-by-cycle computation of ΔK_{th} for a growing crack, with σ^* exhibiting transient response.

The Wheeler and Willenborg [8,9] models approximate retarded crack growth after an overload through a transient retardation factor expressed as a function of crack extension through the overload plastic zone. Their approximation may in fact be traceable to transient variation of ΔK_{th} as the fatigue crack proceeds through the overload plastic zone. Direct measurements of ΔK_{th} transient response after an overload are therefore of practical importance. However, the “no growth” criterion is unsuitable to make such measurements because each ΔK_{th} measurement involves repeat application of overload and also involves substantial crack extension, extending to several times the plastic zone size. Transient measurements demand a process that can associate threshold with the onset of extremely small, but detectable crack extension. If at least a few such readouts can be obtained before the crack tip moves out of the overload transient zone, a meaningful understanding may be obtained of how exactly ΔK_{th} transits from the peak value associated with highest compressive residual stress immediately after the overload to eventually return to baseline magnitude associated with constant amplitude level.

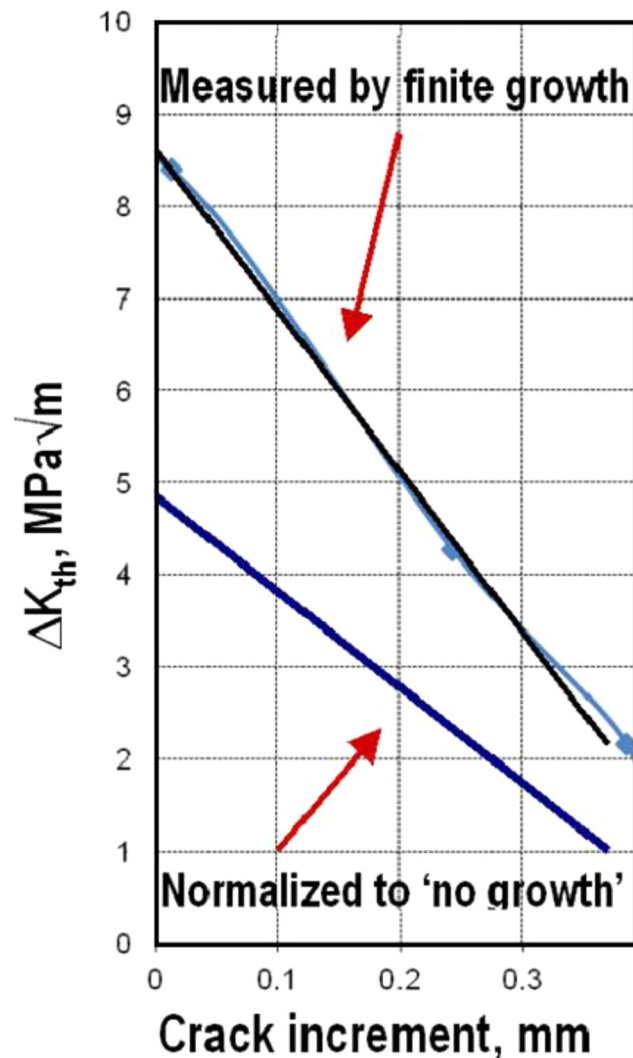
In an attempt to characterize transient ΔK_{th} a pilot test was performed using an “onset of growth” or “finite growth” criterion. This involved 5 % load range *increment* starting from known (conservative) ΔK_{th} value obtained previously under constant amplitude loading with a constant block size of 50 000 cycles until detection of onset of fatigue crack extension exceeding 0.025 mm. This approach is similar to the one adopted by Marci et al. [29] to determine the K_{PR} parameter⁹. We used a certain

⁹It has been shown in previous work [29] that K_{PR} may actually represent the sum $K_{op} + \Delta K_{th}$ that defines the threshold for onset of fatigue crack extension. On the erroneous assumption that $\Delta K_{th} = \text{Const}$, the authors were lead to believe in K_{PR} as a “new” parameter that accounts for both crack closure as well as crack-tip residual stress. The present study, confirms the “hunch” behind the authors’ rationale but demonstrates independent action of two different phenomena operating in synergy.

reference growth rate exceedance given by the ratio $0.025 \text{ mm} / 50\,000 \text{ cycles}$, or, $5 \times 10^{-7} \text{ mm/cycle}$. This ideally allows ΔK_{th} data points to be obtained at intervals that may be as small as 0.025 mm . These data points correspond to a *finite, higher-than-threshold* growth rate and therefore represent *inflated* values of ΔK_{th} . In addition, they may be distorted by the potential contribution to crack extension of $50\,000$ cycle blocks at preceding, lower step value of ΔK . As shown in **Fig. 8**, this procedure does appear to provide a trend line that can be corrected to represent “no growth” conditions. The data in **Fig. 8** illustrate the connection between residual stress distribution in the overload plastic zone and ΔK_{th} , suggesting its linear decrease with transient crack extension. Such data may assist in fine tuning and validating variable amplitude crack growth models.

FIG. 8

Transient ΔK_{th} estimates following overload plotted against crack increment. Normalized straight line appears to suggest linear drop in ΔK_{th} to baseline value as crack tip traverses overload plastic zone.

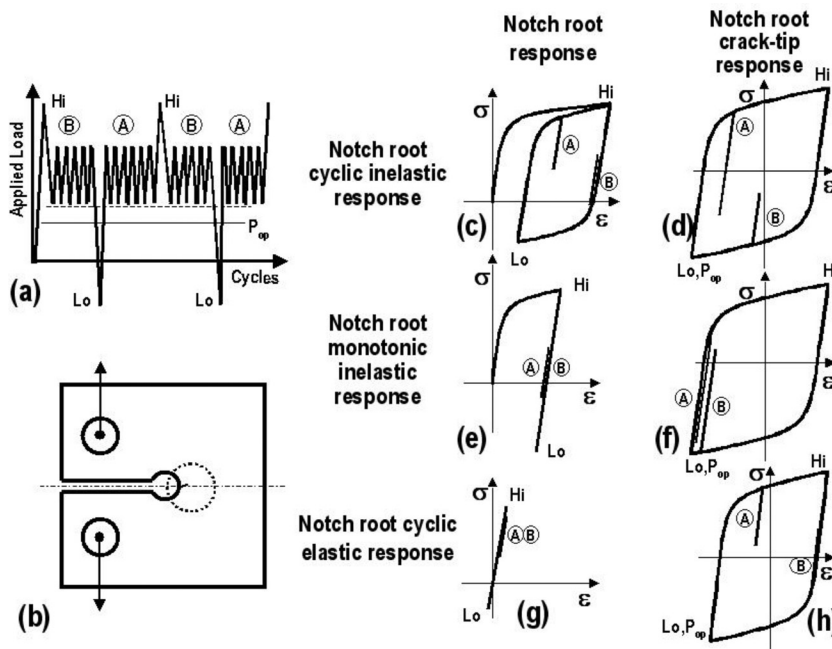


APPLICATION OF RESIDUAL STRESS EFFECT IN FATIGUE ANALYSIS

Local stress-strain analysis (LSS) and fracture mechanics (FM) approaches pursue different paths to characterize fatigue life. If one assumes that fatigue life is predominantly crack growth life, one should expect the two approaches to yield similar results for notch fatigue. **Figure 9** examines the potential for such convergence given a simple variable amplitude load sequence, **9(a)**, acting on a keyhole notched C(T) specimen, **9(b)**. **Figures 9(c), 9(e), and 9(g)** show notch root response and **Figs. 9(d), 9(f), and 9(h)** reflect notch root crack tip response for the three possible levels of notch root response, fully elastic (bottom), monotonic yield (middle), and cyclic inelastic (top). Note that crack tip response for all three cases will be cyclic inelastic. Comparison of the two approaches is based on fatigue damage from identical cycles A and B that lie on opposite sides of tensile and compressive overloads and thereby experience cycle-sequence sensitivity. This is a test of how LSS and FM approaches interpret load sequence sensitivity.

In the event of notch root *cyclic inelastic* response, **Figs. 9(c)** and **9(d)**, the notch root local stress amplitude over both A and B will be identical, but the two will be offset in terms of mean stress. This will lead to greater fatigue damage from A than from B. In the case of the notch root fatigue crack, cyclic notch root inelasticity can cause sequence sensitive change in ΔK_{eff} over cycles A versus B. This is because notch root inelastic response may potentially widen the gap between crack *closure* on the falling half of the major cycle and crack *opening* on the rising half, leading to larger ΔK_{eff} in A than in B [30]. Finally, near-tip mean stress in A exceeds that in B, therefore, cycles A will see reduced ΔK_{th} by comparison to B. Accelerated crack

FIG. 9 Schematic assessment of potential convergence of LSS and FM approaches to notch fatigue under variable-amplitude loading.



growth will follow in cycles A due to both operative mechanisms. One may conclude, that in a qualitative sense, both approaches will indicate similar trends of cycle sequence sensitivity.

In the event of *monotonic* notch root yield (Figs. 9(e) and 9(f)), computed notch root fatigue damage 9(e) will be identical for A and B. There will be no cycle sequence sensitivity though fatigue damage will be reduced in A, B, thanks to drop in local mean stress following the first tensile overload. Notch root crack growth response, Fig. 9(f), will on the other hand see the consequences of both crack closure, as well as crack tip hysteretic response. Thus, if crack closure load is high, cycles A and B will tend towards identical σ^* , leading to equally retarded ΔK_{th} . However, if P_{op} is much lower than P_{min} in these cycles, A and B will see a greater shift in mean stress and consequentially, different σ^* . This can potentially result in vastly different crack growth rates, particularly, if the σ^* values lie in the transient region II in Fig. 5. There is the other possibility that if growth rates are into the Paris Regime, both A and B will see identical growth rates and given absence of crack closure, may not even see retardation. Thus, a fatigue crack growing out of the notch root in this case can see a vast variety of sequence sensitive effects in contrast to what one would expect from traditional notch fatigue analysis. Fractographic evidence from previous work supports these observations [4,5].

Finally, in the event of *elastic* notch root response, Figs. 9(g) and 9(h), notch fatigue analysis will suggest total absence of load interaction effects. However, the notch root fatigue crack will see the same variety of effects as listed for the previous case.

One may conclude that LSS and FM approaches more or less converge only for conditions of cyclic notch root inelasticity. Assuming notch fatigue life is largely determined by notch root crack kinetics, the conventional approach to notch fatigue design may be more limited in its capability to account for load interaction than it would have appeared previously. The prospect of computing notch fatigue life based on the results of this study appears to hold promise. Targeted experiments like in Ref. [5] involving notched coupons subject to simple variable load sequences, combined with analysis using both LSS and FM approaches, are likely to contribute to bridging the divide between the two schools of interpreting fatigue under spectrum loading.

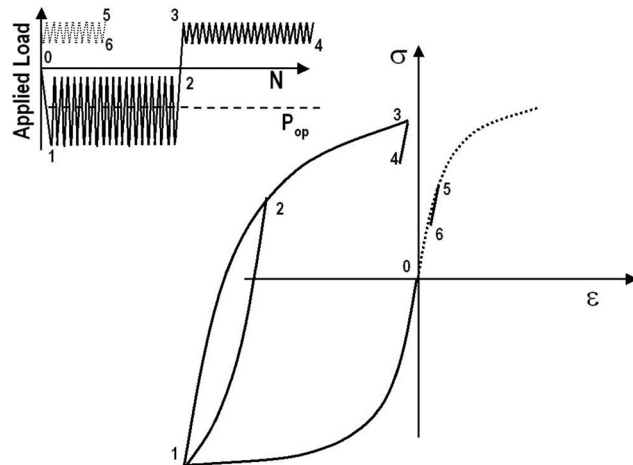
INTERPRETATION OF THE EFFECT OF COMPRESSION PRE-CRACKING ON ΔK_{TH}

Compression pre-cracking has been found to yield more conservative estimates of ΔK_{th} than conventional load shedding scheme per ASTM E647 [1,31]. The summary understanding provided by Fig. 9 of notch root and notch root crack tip response together may explain how and why exactly this occurs.

Figure 10 describes numerically simulated notch root stress-strain response to compression-compression cycling shown at top. At the commencement of loading from “0” to “1,” notch root stress-strain response will follow the monotonic stress strain curve, resulting in potentially substantial yield in compression. This results in local stress excursion into tension during the load excursion 1–2 when the notch root stress-strain response follows the cyclic stress-strain curve. This, combined with near-zero local closure stress caused by notch root cyclic yield, results in notch

FIG. 10

Notch root stress response to compression-compression precracking.



root fatigue crack initiation and growth under the tensile portion of the local stress response 1–2. As the crack tip moves away from the notch root, local tensile stress, “2,” will reduce, and do so faster than stress intensity can increase with crack size. This is the reason for inevitable crack arrest under compression-compression cycling. However, at crack arrest, “2” will still be greater than zero even as maximum load remains compressive. For this reason, when maximum load is switched to “3” for continued tension-tension cycling, local stress will climb even higher due to the excursion 2–3. In fact, it will reach a magnitude the crack tip would not have seen in the absence of compression-compression precracking. Compare 3–4 with 5–6 to note the difference caused by the fact that as opposed to a crack grown under tension-tension cycling, a fatigue crack grown under compression-compression cycling will see a residual tensile stress even when fully unloaded. The application of an incremental tensile load will thus be superposed on such residual notch field stress, leading to higher σ^* . This inevitable incremental tensile crack-tip stress excursion associated with switch over from compression to tension-tension cycling may explain reduced ΔK_{th} after compression-compression precracking, even if closure is not involved.

CONCLUDING REMARKS AND FUTURE WORK

Conventional understanding of cumulative fatigue damage as well as crack growth is built around the perception of metal fatigue as a strain driven phenomenon. However, environmental action including that, associated with near threshold response in atmospheric fatigue, is moderated by stress. The magnitude of stress is defined exclusively by the elastic component of strain, whose variation is hysteretic by nature and therefore not directly amenable to LEFM considerations. The present study accounts for these through the relationship between σ^* and ΔK_{th} . Assuming that near-tip residual stress (or, residual *elastic* strain) controls both diffusion and reaction kinetics, the findings of the study may be extended to model other phenomena that are yet to find robust modeling solutions. These include fatigue stress corrosion cracking and elevated temperature crack growth in creep resistant superalloys,

including the effect of hold time at intermediate load levels, such as in engine load spectra.

Extension of the σ^* concept will demand its adaptation to suit new requirements, hopefully, as a function of LEFM parameters such as K as with governing Eqs 4 and 5 in this study. Thus, if in atmospheric fatigue crack growth, near threshold response is controlled by surface diffusion affecting just a few atomic layers, elevated temperature crack growth will involve oxidation that may run to greater depth beneath the surface. Hold time at elevated temperature will promote intergranular oxidation that can penetrate with time to several grains. Obviously, under such conditions, a different characteristic depth to determine σ^* may be required. Likewise, changing stress state across the thickness will demand consideration of σ^* distribution, rather than a single value, so that crack front shape evolution can be simulated.

Conclusions

1. Threshold stress intensity, ΔK_{th} , in atmospheric fatigue is closely related to load-sequence sensitive near tip residual stress, σ^* , whose characterization demands cycle-by-cycle analysis of near-tip inelastic stress-strain response. An analytical procedure is described to compute σ^* under both constant amplitude as well as simple overload/underload sequences as a cycle-sequence sensitive function of applied loading.
2. A fully automated experimental procedure was developed to determine ΔK_{th} under both constant amplitude loading as well as under periodic overloads. It uses a simple variable-amplitude load sequence involving high stress ratio small amplitude cycles interspersed with periodic tensile and compressive overloads to induce threshold conditions over a wide interval of tightly controlled, reproducible and computable σ^* .
3. Conventional procedures to estimate ΔK_{th} such as ASTM E647 cannot yield useful results for engineering application involving history effects. Experiments on 2024-T3 Al-alloy under a variety of overload/underload conditions show up to a factor of five variation in ΔK_{th} as a function of sequence-sensitive σ^* . Modeling this dependence along with consideration of crack closure is likely to improve our ability to model fatigue crack growth in engineering application.

ACKNOWLEDGMENTS

The experiments were carefully performed by Kiran Kumar as part of his Master's Thesis submitted to Manipal Institute of Technology. The writer is extremely grateful for invaluable suggestions from the reviewers of the original manuscript and for the constructive feedback on this work received after presentations to colleagues at several institutions, notably, Defence Metallurgical Research Laboratories, Hyderabad, Air Force Research Laboratories, WPAFB, OH, Kiev Institute for Problems of Strength, Siberian Aircraft Research Institute, Novosibirsk and Samara Aerospace Institute.

References

- [1] ASTM E647-13: Standard Test Method for Measurement of Fatigue Crack Growth Rates, *Annual Book of ASTM Standards*, ASTM International, West Conshohocken, PA, 2013.

- [2] Ashbaugh, N. E., Porter, W. J., Rosenberger, A. H., and Sunder, R., "Environment-Related Load History Effects in Elevated Temperature Fatigue of a Nickel-Base Super-Alloy," *Proceedings Fatigue*, Stockholm, Sweden, June 2–7, 2002.
- [3] Sunder, R., Porter, W. J., and Ashbaugh, N. E., "The Role of Air in Fatigue Load Interaction," *Fatigue Fract. Eng. Mater. Struct.*, Vol. 26, No. 1, 2003, pp. 1–16.
- [4] Sunder, R., "On the Hysteretic Nature of Variable-Amplitude Fatigue Crack Growth," *Int. J. Fatigue*, Vol. 27, Nos. 10–12, 2005, pp. 1494–1498.
- [5] Sunder, R., "Fractographic Reassessment of the Significance of Fatigue Crack Closure, Fatigue and Fracture Mechanics," *ASTM STP 1461*, S. R. Daniewicz, J. C. Newman and K. H. Schwalbe, Eds., ASTM International, West Conshohocken, PA, 2005, pp. 22–39.
- [6] Sunder, R., "Effect of Periodic Overloads on Threshold Fatigue Crack Growth in Al-Alloys," *Fatigue Fracture Mechanics*, *ASTM STP 1461*, S. R. Daniewicz, J. C. Newman, K. H. Schwalbe, Eds., ASTM International, West Conshohocken, PA, 2005, pp. 557–572.
- [7] Elber, W., "The Significance of Fatigue Crack Closure," *ASTM STP 486*, ASTM International, Philadelphia, PA, 1971, pp. 230–242.
- [8] Wheeler, O. E., "Spectrum Loading and Crack Growth," *ASME J. Basic Eng.*, Vol. 94, 1972, pp. 181–186.
- [9] Willenborg, J., Engle, R. H., and Wood, H. A., "A Crack Growth Retardation Model Based on Effective Stress Concepts," *Report No. AFFDL-TM-71-1 FBR*, Wright-Patterson Air Force Base, OH, 1971.
- [10] Broek, D., *Elementary Engineering Fracture Mechanics*, Martinus Neijhoff Publishers, The Hague, the Netherlands, 1985.
- [11] Anon, "Wöhler's Experiments on the 'Fatigue of Metals'," *Engineering*, Vol. 11, June, 1871, pp. 199–441.
- [12] Bauschinger, J., "On the Change of the Elastic Limit and Strength of Iron and Steel by Tension and Compression, by Heating and Cooling and by Often Repeated Loading," *Technical Report*, Munich Technical University, Munich, Germany, 1886 (in German).
- [13] Ramberg, W. and Osgood, W. R., "Description of Stress–Strain Curves by Three Parameters," *Technical Report No. 902*, National Advisory Committee for Aeronautics, Washington, D.C., 1943.
- [14] Manson, S. S., "Future 'Directions for Low Cycle Fatigue,' Low Cycle Fatigue," *ASTM STP 942*, H. D. Solomon, G. R. Halford and B. N. Leis, Eds., ASTM International, Philadelphia, PA, 1988, pp. 15–39.
- [15] Petit, J., Henaff, G., and Sarrazin-Baudoux, C., "Mechanisms and Modeling of Near-Threshold Fatigue Crack Propagation, Fatigue Crack Growth Thresholds, Endurance Limits and Design," *ASTM STP 1372*, J. C. Newman, Jr. and R. S. Piascik, Eds., ASTM International, West Conshohocken, PA, 2000.
- [16] Sunder, R., "Fatigue as a Process of Brittle Micro-Fracture," *FFEMS*, Vol. 28, No. 3, 2005, pp. 289–300.
- [17] Sunder, R., "A Unified Model of Fatigue Kinetics Based on Crack Driving Force and Material Resistance," *Int. J. Fatigue*, Vol. 29, Nos. 9–11, 2007, pp. 1681–1696.

- [18] Bowles, C. Q., 1978, "The Role of Environment, Frequency and Wave Shape During Fatigue Crack Growth of Aluminum Alloys," Ph.D. thesis, Delft University of Technology, Delft, the Netherlands.
- [19] Gsell, M., Jakob, P., and Menzel, D., "Effect of Substrate Strain on Adsorption," *Science*, Vol. 280, 1998, pp. 717–720.
- [20] Neuber, H., 1937, *Kerbspannungslehre [Theory of Notch Stresses]*, J. W. Edwards, Ed., Springer, New York, 1946.
- [21] SAE, *SAE Fatigue Design Handbook*, 3rd ed., Society for Automotive Engineers, Warrendale, PA, 1997.
- [22] Wetzel, R. M., 1971, "A Method for Fatigue Damage Analysis," Ph.D. thesis, Department of Civil Engineering, University of Waterloo, Waterloo, ON, Canada.
- [23] Marci, G., "Non-Propagation Conditions (ΔK_{th}) and Fatigue Crack Propagation Threshold (ΔK_T)," *Fatigue Fract. Eng. Mater. Struct.*, Vol. 17, No. 8, 1994, pp. 891–907.
- [24] Bradshaw, F. J. and Wheeler, C., "The Effect of Gaseous Environment and Fatigue Frequency on the Growth of Fatigue Cracks in Some Aluminium Alloys," *Int. J Fract. Mech.*, Vol. 6, 1969, pp. 255–268.
- [25] Ashbaugh, N. E., Dattaguru, B., Khobaib, M., Nicholas, T., Prakash, R. V., Ramamurthy, T. S., and Seshadri, B. R., "Experimental and Analytical Estimates of Fatigue Crack Closure in an Aluminum–Copper Alloy. Part I. Laser Interferometry and Electron Fractography," *Fatigue Fract. Eng. Mater. Struct.*, Vol. 20, No. 7, 1997, pp. 951–961.
- [26] Vasudevan, A. K., Sadananda, K., and Louat, N., "A Review of Crack Closure, Fatigue Crack Threshold and Related Phenomena," *Mater. Sci. Eng.*, Vol. 188, Nos. 1–2, 1994, pp. 1–22.
- [27] Hopkins, S. W., Rau, C. A., Leverant, G. R., and Yuen, A., "Effect of Various Programmed Overloads on the Threshold for High-Frequency Fatigue Crack Growth, Fatigue Crack Growth Under Spectrum Loads," *ASTM STP 595*, ASTM International, Philadelphia, PA, 1976, pp. 125–141.
- [28] Sunder, R., "Unraveling the Science of Variable Amplitude Fatigue, Fatigue and Fracture Mechanics: 38th Volume," *ASTM STP 1546*, ASTM International, West Conshohocken, PA, 2012, pp 20–64.
- [29] Lang, M., "A Model for Fatigue Crack Growth, Part I: Phenomenology," *Fatigue Fract. Eng. Mater. Struct.*, Vol. 23, No. 7, 2000, pp. 587–601.
- [30] Sunder, R., "Notch Root Crack Closure Under Cyclic Inelasticity," *Fatigue Fract. Eng. Mater. Struct.*, Vol. 16, No. 7, 1993, pp. 677–692.
- [31] Newman, J. C., Jr., Ziegler, B. M., Shaw, J. W., Cordes, T. S., and Lingenfelter, D. J., "Fatigue Crack Growth Rate Behavior of A36 Steel Using ASTM Load-Reduction and Compression Precracking Test Methods," *J. ASTM Int.*, Vol. 9, No. 4, JAI103966.